Solving the Model Gap — Executive Summary

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AMD-Sourcing the Engineering Cycles to Close NVIDIA's Physical-AI Lead

AMD hardware already wins on economics — roughly 50% lower capex and 30% lower opex than the incumbent, on competitive silicon (MI300X, Ryzen AI, Kria). Yet application developers still say no, because their products are already built around NVIDIA's models and pipelines (NGC, TensorRT, Isaac ROS, cuRobo, Isaac Sim, Cosmos). NVIDIA's moat is roughly 80% ecosystem and 20% hardware: 2M+ robotics developers, a 7+-year head start. Re-hosting a deployment costs 6–18 engineer-months and 5–7 re-targeted artifacts per model — a bill the customer, not AMD, has to pay before hardware economics even enter the conversation. That burden falls hardest on the small and mid-size robotics companies AMD most needs to win. Internally, this gap is known and winnable but cannot be closed by any single business unit: model training is iterative, the payoff is multi-year, and no BU can book near-term ROI on it — so, rationally, nobody starts, and the gap widens.

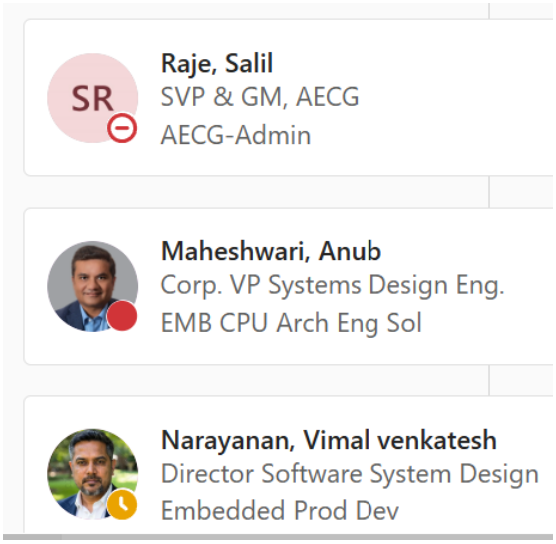
The way out is to pose this as a *replication problem, not an invention problem*. The target models are named, mostly open, and their performance is already published on NGC — we know exactly what "done" looks like. Only two things are missing: compute, which AMD already has (ALOLA and other AMD-wide clusters), and data, the real work (synthetic generation, pooled and curated open data, semi-supervised labeling, pseudo-labels from large open-source models and data pools). The proposal is an AMD-wide model training challenge: pick a target model, train it on AMD compute, submit a per-model leaderboard, and have PAVS validate and publish the result. This pattern has closed comparable capability gaps before — ImageNet seeded the deep-learning era, the DARPA Grand Challenge seeded self-driving, and Kaggle/Netflix Prize crowdsourced solutions that beat in-house teams at a fraction of the cost. A scored, public target repeatedly does what no single roadmap can.

The mechanism is self-sustaining because every party wins by playing: participating engineers get recognition, a publishable track record, and a countable contribution to AMD; PAVS becomes the customer gateway for physical AI and the bridge across AMD's distributed teams; and AMD closes a strategic ecosystem gap with pooled compute and no single-BU budget line, building a pipeline of market-ready, AMD-native models and datasets along the way. The prize is worth the effort — AMD projects a $200B physical-AI silicon market by 2035, robotics startup funding hit $22.2B in 2025 (+69% YoY). Of 125 NGC models scanned, 48 across four priority tiers need an AMD-native replica (Appendix, Tables I–IV); Tier 1 + Tier 2 (30 models) is the actual "customer has this running on NVIDIA today and can't move it" set.

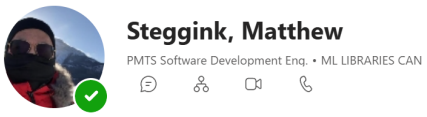
The Ask

The plan is already staffed — we only need the mandate to start.

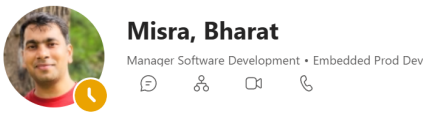
What we need from leadership: an org-wide email endorsing the challenge — the signal that gives the effort good traction.



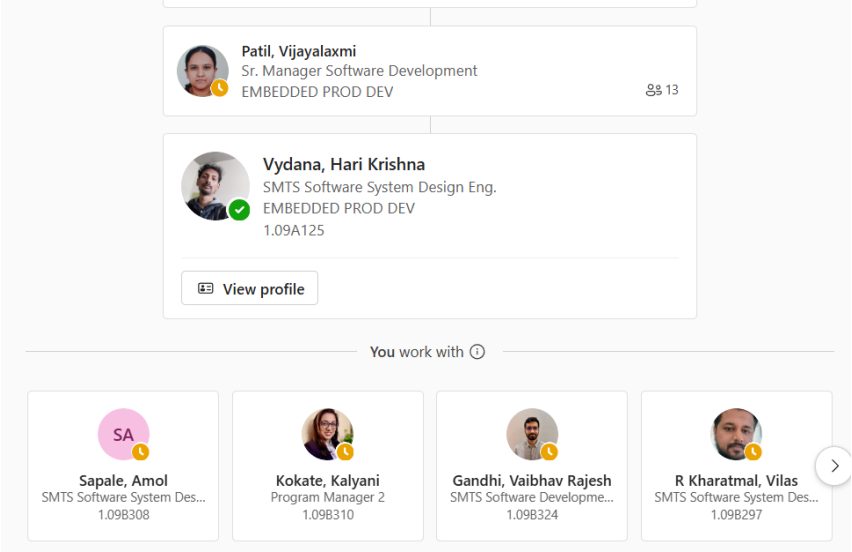
Leadership endorsement



Compute (ALOLA)



Per-model GitHub



Team to execute

Fig. 1. Already staffed and committed: leadership signal, compute, repos, and team are all in place — all that's missing is the go-ahead.

APPENDIX A
NGC PHYSICAL-AI TARGET SET

TABLE I
TIER 1 — PRODUCTION PERCEPTION STACK (NVIDIA METROPOLIS/DEEPSTREAM). HIGHEST PRIORITY — MOST COMMONLY DEPLOYED NGC MODELS A CUSTOMER RUNS TODAY AND CANNOT CARRY OVER TO AMD.

NVIDIA model	Function	Why it matters
TrafficCamNet	Vehicle detection (traffic cameras)	Core Metropolis/smart-city SKU
PeopleNet / PeopleNet-AMR	Person detection	Most widely deployed DeepStream model
LPRNet + LPDNet	Plate detection + recognition pair	Sold as a pair, tolling/parking/security
VehicleMakeNet / VehicleTypeNet	Vehicle attribute classification	Paired with TrafficCamNet in the same pipelines
ReIdentificationNet (+ Transformer)	Person re-identification across cameras	Retail/security analytics
PeopleSemSegNet / AMR	Person segmentation	Privacy-masking, robotics safety zones
Retail Object Detection / Recognition	Shelf/product detection	Retail analytics vertical
Action Recognition Net	Action/behavior classification	Security, workplace safety
BodyPoseNet / PoseClassificationNet	Human pose estimation	Robotics, ergonomics, fitness
OCDNet / OCRNet	Scene text OCR	Logistics, document/label reading

TABLE II
TIER 2 — ROBOTICS/PERCEPTION BUILDING BLOCKS (PHYSICAL AI). THE MODELS ISAAC ROS PERCEPTION PIPELINES ARE BUILT ON — CLOSEST TO AMD-ROS/AARTS. TIER 1 + TIER 2 = 30 MODELS, THE REAL "CUSTOMER HAS THIS RUNNING AND CAN'T MOVE IT" SET.

NVIDIA model	Function
Bi3D Proximity Segmentation	Stereo-based obstacle proximity
ESS DNN Stereo Disparity	Stereo depth
FoundationPose	6-DoF object pose estimation/tracking
PointPillarNet	LiDAR 3D object detection
Transfusion (3D Object Detection)	Camera+LiDAR fusion detection
CitySemSegFormer	Semantic segmentation (autonomy/city-scale)
TAO Pretrained (Classification/DetectNet V2/EfficientDet/Instance Seg/Semantic Seg)	Generic backbones customers fine-tune on top of — AMD needs some pretrained backbone family here or customers cannot start a TAO-style workflow on AMD

TABLE III
TIER 3 — BIOMETRIC/HUMAN-SENSING (SMALLER BUT STICKY NICHE). MATTERS MAINLY IF AMD IS ALSO CHASING THE BIOMETRIC/KIOSK VERTICAL.

NVIDIA model	Function
FaceNet	Face detection
FaceDetectIR	Face detection (infrared)
GazeNet	Gaze tracking
EmotionNet	Emotion classification
GestureNet	Hand/body gesture recognition
HeartRateNet	Remote heart-rate estimation
Facial Landmark Estimator (FPENet)	Facial landmark detection
Deformable DETR (NvImageNet weights)	Detection backbone
FAN (NVImageNet weights)	Classification backbone
GCViT (NVImageNet weights)	Classification backbone
Visual ChangeNet Classification	Change detection classification
NVIDIA Optical Inspection Model	Industrial visual inspection

TABLE IV
TIER 4 — ENTERPRISE VISION & HPC/CLIMATE (LOWEST PRIORITY). MATTERS MAINLY IF AMD IS ALSO CHASING THE ENTERPRISE-VISION OR HPC-CLIMATE VERTICAL.

NVIDIA model	Function	Why lower priority
PCB Defect Classification	Industrial inspection	NVIDIA AI Enterprise EULA; small footprint
PeopleSemSegFormer	Person segmentation	NVIDIA AI Enterprise EULA; small footprint
DINO (NvImageNet weights)	Classification backbone	NVIDIA AI Enterprise EULA; small footprint
PhysicsNeMo: AFNO_DX_FI-V1-ERA5	Weather forecasting	Community Model License; HPC/climate-only
PhysicsNeMo: AFNO_DX_SR-V1-ERA5	Weather forecasting	Community Model License; HPC/climate-only
PhysicsNeMo: AFNO_DX_TP-V1-ERA5	Weather forecasting	Community Model License; HPC/climate-only
PhysicsNeMo: AFNO_DX_WG-V1-ERA5	Weather forecasting	Community Model License; HPC/climate-only
PhysicsNeMo: DLESyM-V1-ERA5	Weather forecasting	Community Model License; HPC/climate-only

48 models across four tiers, out of 125 scanned on NGC — the other 77 are either already portable (49) or unresolved catalog entries (28).